Notice that here we have *surfaces* of discontinuity, where the cost function changes its value: when `|x|==|y|` (v lying on a line passing through the diagonals of the square), and when `x` or `y` are equal to `0` or `100` (lines lying on the perimeter of the square). For any point on these surfaces, the cost will be infinite, because, as shown in the third example in the figure, vertex v will lie on one of the edges between the other vertices (and our constraints assign an infinite cost to these invalid configurations).

Moreover, as you can see, no matter how hard we try, we can't find a position for vertex v that will guarantee us an embedding with the minimum possible crossing number: this is because the position of the other 4 vertices is not optimal for a straight-line drawing of K_5.

In turn, this is because, if we consider the larger problem of finding the best embedding (with none of the vertices locked), we hit a **local minimum** of the cost function: a point or region in the n-dimensional problem space where the cost function has a lower value than in the surrounding area, but not the lowest possible overall.

If we could take a 2D projection of the cost function on a generic configuration of the graph (not specifically the embedding in figure 16.8), it might hypothetically look somehow similar to figure 16.9.